Data Analytics Capstone

Using Machine Learning to Predict Complaint Resolution Outcomes in Retail Banking: Evidence from Open Consumer Financial Complaint Data

Detailed Interim Report

Zahoor ul Islam

Walsh College

QM640: Data Analytics Capstone

Jainesh Garg

Summer 2026 Term

July 27, 2026

GitHub Repository

The project repository containing the data-processing workflow, notebooks, outputs, and report artifacts is available at: https://github.com/zahoorulislam/retail-complaint-resolution-ml

The repository structure follows the five-notebook workflow used in this report: data extraction, exploratory analysis, hypothesis testing, model training, and model interpretation. To keep the main report focused on decision-ready findings, detailed secondary tables and supporting figures are moved to the appendices and referenced from the relevant sections.

Introduction

Complaint management is a critical retail banking function because complaints reveal points where product design, digital servicing, customer communication, regulatory obligations, and customer trust intersect. In a high-volume environment, complaint teams need more than descriptive reports after complaints are closed. They need early-warning signals that help identify which complaints may require faster review, specialist handling, or escalation.

This capstone project uses open Consumer Financial Protection Bureau (CFPB) complaint data to examine and predict complaint resolution outcomes in retail banking. The analysis combines structured fields such as product, issue, company, state, submission channel, and complaint year with narrative-related features. The project is designed as a decision-support study rather than an automated complaint-decision system.

The managerial value of the project is that it converts a large public complaint dataset into operational insight. The results can inform product-level monitoring, issue routing, capacity planning, early-warning dashboards, threshold-based review queues, and responsible model governance. The technical value is that it compares baseline, logistic regression, text-enhanced logistic regression, and random forest models under severe class imbalance.

Background and Context

Retail banking complaints can involve checking accounts, cards, mortgages, prepaid cards, money transfers, vehicle finance, consumer loans, and digital wallet or payment services. These complaints vary in complexity. Some can be handled through standard explanations, while others may require investigation, remediation, monetary or non-monetary relief, or escalation to specialized teams.

The CFPB Consumer Complaint Database is a public source of consumer complaints sent to companies for response. It includes structured complaint descriptors, company response fields, timeliness information, and, where available, consumer narratives. Because the data include both structured and text-based attributes, they provide a useful setting for testing whether machine learning can support complaint risk prediction and operational analysis (Consumer Financial Protection Bureau, n.d.-a; CFPB Open Tech, n.d.).

The project is relevant to multiple stakeholders. Complaint operations teams can use risk signals for triage and workload planning. Product owners can use product- and issue-level patterns to review operational pain points. Risk and compliance teams can use subgroup and threshold monitoring to govern model use. Analytics teams can use the notebooks and outputs as a reproducible framework for future internal complaint modeling.

Problem Statement

Retail banking complaint teams often rely on manual review, rule-based routing, and historical reporting. These approaches are useful, but they may not identify delayed-response risk early enough, and they may not show which combinations of product, issue, company, narrative, geography, and time signals create higher operational complexity. The gap addressed by this project is the lack of a reproducible, data-driven framework for predicting complaint response timeliness and explaining business-relevant complaint outcome patterns using open complaint data.

Purpose of the Study

The purpose of the study is to classify and explain complaint resolution outcomes in retail banking. Specifically, the study predicts response timeliness, examines company response outcomes, tests whether customer narratives add predictive value, and evaluates whether company and geography variables add incremental signal beyond complaint characteristics. The intended output is a business-oriented analytical framework that supports triage, monitoring, governance, and future model improvement.

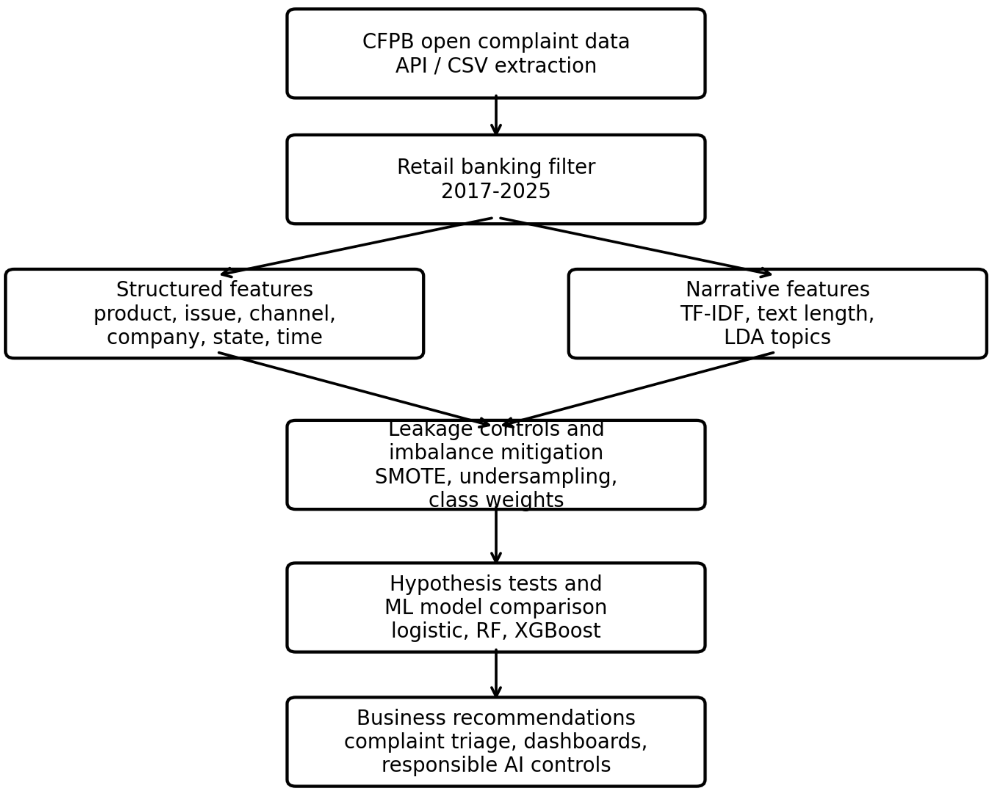
Interim Project Status (Progress Snapshot)

- Completed: Raw CFPB CSV validation, chunk-based extraction, product and date filtering, duplicate review, feature creation, and data dictionary preparation.
- Completed: Exploratory data analysis covering product, issue, company, state, submission channel, timely response, company response, narrative availability, time trends, process lag, and tags.
- Completed: Hypothesis testing for product, issue, company, and state relationships with complaint outcomes.
- Completed: Machine learning comparison using a dummy baseline, structured logistic regression, structured plus TF-IDF logistic regression, and structured random forest.
- Completed: Model interpretation, threshold analysis, error analysis, subgroup performance review, feature-family importance, limitations, and implementation roadmap.

- Updated in this version: More detailed business outcomes have been added, and secondary tables and supporting figures have been moved to the appendices for cleaner executive readability.
- Pending: Final repository cleanup, mentor review, and final submission polish.

Figure 1

Project workflow for complaint-resolution analytics.



Note. The workflow connects the five uploaded notebooks: extraction, EDA, hypothesis testing, model training, and interpretation. Detailed notebook evidence is provided in Appendix D.

Scope and Objectives

The scope is limited to retail banking-related CFPB complaint records from 2017 through 2025. The product filter includes checking and savings accounts, bank account or service, credit cards and prepaid cards, mortgages, vehicle loans or leases, consumer loans, payday/title/personal loans, money transfers, virtual currency, and related money-service products. The unit of analysis is one complaint record.

The objectives are to prepare a clean and reproducible dataset; describe complaint patterns through exploratory analysis; test relationships between complaint attributes and outcomes; compare predictive models under class imbalance; interpret model behavior; and translate the findings into business outcomes, governance controls, limitations, and a practical implementation roadmap.

Table 1

Research questions, hypotheses, and expected business outcomes.

RQ	Research Question	Hypothesis	Business Outcome
RQ1	Does product type significantly influence timely complaint response?	Product type is associated with response timeliness.	Product-level monitoring, staffing review, and workflow redesign.
RQ2	Do issue categories predict different company response outcomes?	Issue category is associated with company response outcome.	Routing improvement, specialist assignment, and root-cause analysis.
RQ3	Do customer narratives improve prediction beyond structured fields?	Narrative features improve some predictive metrics.	NLP-supported triage and complexity detection.
RQ4	Do company and geography add value beyond complaint characteristics?	Company/state features add incremental predictive signal.	Governance monitoring, benchmarking, and regional review.

Literature Survey

Literature Review Approach

The literature review was organized around five themes: complaint handling and service recovery, predictive modeling in financial complaints, natural language processing for complaint narratives, class imbalance and evaluation metrics, and responsible AI governance. Sources were selected because they support one or more research questions and justify the chosen modeling decisions, including logistic regression, random forests, text features, leakage control, threshold evaluation, and fairness monitoring.

Table 2

Full literature relevance matrix.

Author / Source	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports the Project / RQ Linkage
Consumer Financial Protection Bureau (n.d.-a)	Consumer complaints and financial services oversight	Public Consumer Complaint Database	Public administrative complaint dataset and complaint metadata	Provides complaint records, products, issues, company responses, timeliness, channels, geography, and narrative availability.	Primary data source for all research questions; supports product, issue, company, geography, narrative, and outcome analysis.
Consumer Financial Protection Bureau (n.d.-b)	Public complaint-data sharing and privacy	CFPB data-sharing guidance	Policy and disclosure documentation	Explains that public complaint information is shared with limits and that narratives depend on consumer consent and disclosure rules.	Supports limitations, narrative availability discussion, privacy controls, and responsible use of consumer text data in RQ3.
CFPB Open Tech (n.d.)	Complaint database structure	CFPB API field reference	Technical field documentation	Defines complaint fields and variable meanings, including product, issue, company response, submitted via, and timely response.	Supports data dictionary, feature definitions, leakage review, and reproducibility across all notebooks.
Data.gov (n.d.)	Open government data availability	Consumer Complaint Database catalog record	Data catalog and metadata documentation	Confirms public availability of the dataset and provides access context for evaluator reproducibility.	Supports GitHub data availability statement, source transparency, and project reproducibility requirements.
Tax et al. (1998)	Service recovery and complaint handling	Customer complaint experiences	Conceptual and empirical service recovery analysis	Complaint experiences shape customer evaluations, relationship outcomes, and perceptions of fairness.	Supports the business rationale that complaint response quality matters for customer trust and operational performance.
Maxham and Netemeyer (2002)	Complaint handling over time	Customer perceptions of complaint handling	Longitudinal modeling of service recovery perceptions	Perceived justice and complaint handling influence satisfaction and future customer intentions.	Supports the operational importance of timely and appropriate complaint handling in the introduction and RQ2.
Homburg and Fürst (2005)	Organizational complaint handling	Complaint-management practices	Analysis of mechanistic and organic complaint-handling approaches	Complaint-handling design can affect customer loyalty and should combine process discipline with flexibility.	Supports recommendations for workflow redesign, escalation controls, and specialist handling in RQ1 and RQ2.
Hosmer et al. (2013)	Logistic regression and classification	Applied binary-outcome modeling	Logistic regression theory and applied methods	Logistic regression is useful for interpretable classification and estimating direction of associations.	Justifies structured logistic regression as an interpretable baseline for timely-response prediction and coefficient interpretation.
Breiman (2001)	Ensemble learning	General supervised learning settings	Random forest methodology	Random forests improve predictive performance by combining many decision trees and capturing nonlinear patterns.	Justifies the structured random forest model used as the best-performing ROC-AUC model in RQ3/RQ4.
Pedregosa et al. (2011)	Machine-learning implementation	Python scikit-learn ecosystem	Open-source machine-learning library documentation and paper	Scikit-learn provides reliable implementations for preprocessing, model training, pipelines, and evaluation.	Supports the technical approach used in the notebooks for TF-IDF, logistic regression, random forest, metrics, and validation.
Blei et al. (2003)	Text mining and topic modeling	Document collections	Latent Dirichlet allocation	Text data can reveal hidden themes and structure in large document collections.	Supports the narrative-analysis logic and the broader use of text features for complaint understanding in RQ3.
Loughran and McDonald (2011)	Financial text analytics	Financial disclosures	Textual analysis and domain-specific language evaluation	Words can have domain-specific meaning in financial settings, so text interpretation requires context.	Supports cautious interpretation of TF-IDF terms in complaint narratives and privacy-aware NLP governance.
Chawla et al. (2002)	Class imbalance	Imbalanced classification problems	SMOTE oversampling method	Rare classes require special treatment because standard classifiers may favor the majority class.	Supports the class-imbalance discussion and explains why metrics beyond accuracy and resampling options must be considered.
Davis and Goadrich (2006)	Model evaluation under imbalance	Binary classifiers and ranking metrics	ROC and precision-recall analysis	Precision-recall analysis is important when the positive class is rare and ROC results alone may not show alert quality.	Supports use of ROC-AUC, average precision, classwise precision/recall, and threshold analysis for non-timely risk detection.

Author / Source	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it Supports the Project / RQ Linkage
Kaufman et al. (2012)	Data leakage in predictive modeling	Data-mining workflows	Leakage formulation and prevention guidance	Predictive features must reflect information available at prediction time; post-outcome fields can inflate performance.	Supports exclusion of post-response fields such as company response and public response from intake-stage prediction.
Ribeiro et al. (2016)	Explainable machine learning	Classifier explanations	Local interpretable model-agnostic explanations	Model explanations improve trust by showing which features influence predictions.	Supports feature-importance review, coefficient interpretation, and decision-support use instead of black-box automation.
Mehrabi et al. (2021)	Bias and fairness in machine learning	Machine-learning systems	Survey of bias and fairness concepts	Bias can enter through data, labels, features, measurement, and deployment context.	Supports subgroup performance review by product, state, company, channel, year, and customer tags.
Barocas et al. (2023)	Responsible AI and fairness	Machine-learning governance and fairness	Conceptual and methodological treatment of fairness	Fairness requires careful attention to measurement, proxy variables, institutional context, and human oversight.	Supports governance recommendations, human-in-the-loop use, privacy review, and avoidance of direct company/state ranking.
Vaishnav et al. (2024)	Machine learning on CFPB complaints	CFPB complaint data	Predictive analysis using machine-learning methods	Shows that CFPB complaint records can support supervised predictive analysis and operational insight.	Directly supports the feasibility of applying ML to CFPB complaint data and links to all RQs.

The literature supports a combined analytical design. Complaint-management studies justify the business need for improving complaint handling and service recovery. Predictive modeling literature justifies comparing interpretable and nonlinear classifiers. Imbalance and evaluation literature explains why ROC-AUC, balanced accuracy, precision, recall, F1-score, and threshold analysis are more useful than accuracy alone. Responsible AI literature supports subgroup monitoring, leakage prevention, and human-in-the-loop deployment.

Data Description

Data Source(s) and Access

The primary data source is the CFPB Consumer Complaint Database. The report uses a local downloaded CSV file processed through Notebook 01. The repository is structured so raw data, processed data, code, tables, figures, and report artifacts can be reproduced from the notebooks (Consumer Financial Protection Bureau, n.d.-a; Data.gov, n.d.).

Table 3

Dataset overview after extraction and filtering.

Item	Value
Raw source	CFPB Consumer Complaint Database CSV
Processed records	1,289,027
Processed columns	25
Study period	2017-01-01 to 2025-12-31
Unit of analysis	One CFPB complaint record
Primary target	timely_response_binary: 1 = timely, 0 = non-timely
Secondary target	company_response_group: closed with explanation, relief provided, other response, in progress

Item	Value
Narrative feature	Narrative availability and narrative length; TF-IDF used in a separate narrative model
Main model sample	150,000-record stratified modeling sample

The processed dataset contains 1,289,027 retail banking-related complaint records and 25 columns. Timely responses dominate the dataset, creating a highly imbalanced target. Detailed variables and handling notes are moved to Appendix A to reduce main-body table density while still meeting the data dictionary requirement.

GitHub Data Availability Statement

The raw data are expected under data/raw/complaints.csv. The processed dataset is saved under data/processed/cfpb_retail_complaints_2017_2025.csv. The code notebooks are stored in the notebooks folder, and generated tables, figures, and model artifacts are stored under outputs/tables, outputs/figures, and outputs/models. The appendix includes a notebook workflow evidence table that maps each notebook to its contribution.

Analysis

Data Cleaning

Notebook 01 used chunk-based processing to handle the large raw CSV file. The cleaning process inspected raw headers, selected required fields, standardized column names, parsed complaint dates, applied the 2017-2025 date filter, filtered to relevant retail banking products, created engineered time and narrative features, and saved a processed dataset and sample file. Leakage-sensitive fields were excluded from intake-stage timely-response modeling where they would not be available before a company response.

Table 4

Data cleaning and preparation summary.

Issue	Variables Affected	Treatment Applied	Business Rationale
Large raw file	All fields	Read in chunks and selected required fields	Made processing feasible and reproducible.
Date validity	Date received, date sent to company	Parsed with safe datetime conversion	Enabled period filtering and process-lag features.

Issue	Variables Affected	Treatment Applied	Business Rationale
Product scope	Product	Filtered to retail banking-related products	Aligned dataset with project scope.
Target creation	Timely response?	Mapped Yes/No to binary target	Supported classification modeling.
Narrative features	Consumer complaint narrative	Created availability and length fields; text used in TF-IDF model	Supported RQ3.
Leakage control	Company response/public response fields	Excluded from timely-response prediction	Avoided using post-outcome information.

EDA Results (Interpretation Emphasis)

The exploratory analysis showed that complaint volume is concentrated in a limited number of retail banking products and companies, while outcome variables are highly imbalanced. The timely-response target is especially imbalanced, with the non-timely class representing a small minority of records. This directly shaped the evaluation plan and created the need for classwise metrics and threshold analysis.

Table 5

EDA insight summary and business implications.

EDA Area	Key Insight	Business Meaning
Product distribution	Complaint volume is concentrated in major retail banking products.	Supports product-level monitoring and RQ1.
Timely-response distribution	Most complaints are timely, while non-timely cases are rare.	Accuracy alone is not sufficient for model evaluation.
Company response distribution	Closed with explanation dominates; relief provided is the main secondary outcome.	Supports response outcome modeling and relief triage.
Narrative availability	Only part of the dataset includes consumer narratives.	NLP findings should be treated as supporting signals.
Time trends	Complaint patterns and model performance vary by year.	Supports drift monitoring and retraining.

Modelling

Choice of Models with Justification

The modeling strategy compared a dummy baseline, structured logistic regression, structured plus TF-IDF logistic regression, and structured random forest. The dummy baseline demonstrated why accuracy is misleading. Logistic regression provided interpretability and coefficient direction. TF-IDF logistic regression tested the value of customer narratives. Random forest captured nonlinear and interaction effects across product, issue, company, state, time, and narrative-derived fields (Breiman, 2001; Hosmer et al., 2013; Pedregosa et al., 2011).

Table 6

Choice of models and business justification.

Model	Feature Set	Analytical Purpose	Business Value
Dummy baseline	No predictive features	Majority-class reference point	Shows whether performance is only due to class imbalance.
Structured logistic regression	Product, issue, company, state, channel, tags, year, narrative length	Interpretable benchmark	Provides directional coefficients and governance-friendly explanation.
Structured + TF-IDF logistic regression	Structured features plus narrative text terms	Narrative value test	Tests whether consumer narrative content improves prediction.
Structured random forest	Structured operational features	Final selected model by ROC-AUC	Captures nonlinear and interaction effects for risk triage.

Features Included and Feature Engineering

The final modeling features included product and sub-product, issue and sub-issue, company group, state group, submission channel, customer tags, year received, narrative availability, and narrative length. A separate TF-IDF model used narrative text terms to evaluate the incremental value of customer-written complaint narratives. Post-response fields were excluded from timely-response prediction to avoid leakage.

Evaluation Metrics (Formulae and Calculations)

Table 7

Evaluation metrics and formulae.

Metric	Formula	Interpretation	Used For
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall correct predictions; limited value under imbalance.	Reference only
Balanced accuracy	(Sensitivity + Specificity) / 2	Average performance across both classes.	Model comparison
Precision	TP / (TP + FP)	Share of predicted positives that are correct.	False-alert burden
Recall	TP / (TP + FN)	Share of actual positives captured.	Missed-risk control
F1-score	2 × (Precision × Recall) / (Precision + Recall)	Balances precision and recall.	Classwise assessment
ROC-AUC	Area under ROC curve	Discrimination across thresholds.	Primary model selection
Average precision / PR-AUC	Area under precision-recall curve	Minority-class ranking quality.	Risk-class evaluation

For the selected random forest at the default threshold, the confusion matrix contained 514 correctly identified non-timely complaints, 119 missed non-timely complaints, 34,964 correctly identified timely complaints, and 9,403 timely complaints flagged as non-timely risk.

This shows a conservative risk-detection pattern: the model catches many risk cases but creates a material false-alert workload.

Class Imbalance Treatment

Class imbalance was a central modeling issue because timely responses dominated the target variable. In the final modeling sample, approximately 98.59% of complaints were timely and only 1.41% were non-timely. This distribution means that a simple majority-class model can appear highly accurate while providing no useful ability to identify the rare non-timely complaints that matter most for operational risk management.

The analysis treated this imbalance in several ways. First, the train-test split was stratified so that the rare non-timely class remained represented in both training and testing data. Second, class-weighted models were used where appropriate so that the learner did not treat the majority timely class as the only important outcome. Third, the study reported balanced accuracy, ROC-AUC, average precision, classwise precision, classwise recall, F1-score, and confusion matrices rather than relying on accuracy alone. These metrics made it possible to evaluate whether the model was actually detecting non-timely risk instead of simply predicting most complaints as timely.

The project also used threshold analysis to translate the model into a practical triage setting. Lower thresholds identify more potential delayed-response cases but create a larger false-alert workload. Higher thresholds create smaller, higher-confidence review queues but miss more actual non-timely complaints. Resampling methods such as SMOTE were reviewed conceptually; however, the main pipeline preserved the observed complaint distribution because the data contained high-cardinality categorical features and narrative text, where synthetic records could be difficult to interpret operationally. The final recommendation is therefore to use class-weighted learning, imbalance-sensitive metrics, threshold governance, and human review rather than accuracy-based automation.

Preliminary Results

Findings by Research Question

RQ1 found that product type is statistically associated with timely response, but the practical effect size is very weak. This means product is a useful operational signal but not a complete explanation of delayed response. In business terms, product-level monitoring can identify where workflow review or specialized handling may be useful, but product alone should not drive escalation decisions.

RQ2 found that issue category is associated with company response outcome. This result supports issue-level routing and root-cause analysis. Some issues are more likely to result in relief or other outcomes, while others are more likely to close with explanation. Complaint operations can use this insight to develop issue-specific response playbooks and specialist assignment rules.

RQ3 showed partial support for narrative value. The structured plus TF-IDF model improved accuracy, timely recall, and timely F1-score, but it reduced balanced accuracy, non-timely recall, and non-timely ROC-AUC compared with the structured logistic model. The business conclusion is that narratives add context, but text features should be used as supporting signals rather than as a standalone risk engine.

RQ4 provided the clearest incremental result. Adding company and state features improved balanced accuracy by 0.1418 and ROC-AUC by 0.1479 compared with complaint characteristics alone. This indicates that company and geography contain useful operational signals. However, these variables should not be treated as direct rankings because market share, complaint volume, product mix, customer base, servicing model, and case complexity are not fully controlled.

Table 8

Primary hypothesis and model evidence summary.

RQ	Evidence/Test	Effect or Difference	Decision/Interpretation
RQ1	Chi-square: product vs. timely response	Cramer's V = 0.0928	Reject H0. Product is statistically associated with timeliness, but the practical effect is very weak.
RQ2	Chi-square: issue group vs. company response	Cramer's V = 0.1874	Reject H0. Issue category is associated with response outcome and supports routing/root-cause review.
RQ3	Structured model vs. structured + TF-IDF	Accuracy +0.0779; non-timely recall -0.1864	Partial support. Narratives add context but do not improve every risk-sensitive metric.
RQ4	Base complaint model vs. company/state extended model	Balanced accuracy +0.1418; ROC-AUC +0.1479	Supported. Company and state add predictive value beyond complaint characteristics.

Preliminary Model Performance

Table 9

Primary model performance comparison.

Model	Accuracy	Balanced Accuracy	F1	ROC-AUC	Business Interpretation
Structured random forest	0.7884	0.8000	0.8802	0.8748	Best ROC-AUC and strongest balanced risk-detection profile.
Structured logistic regression	0.7333	0.7908	0.8440	0.8506	Strong interpretable benchmark with useful coefficient direction.
Structured + TF-IDF logistic regression	0.8112	0.7384	0.8947	0.8357	Improves overall accuracy and timely F1; mixed for non-timely risk.
Dummy baseline	0.9859	0.5000	0.9929	0.5000	Reference only; high accuracy is caused by class imbalance.

The structured random forest was selected as the preferred model because it produced the strongest ROC-AUC and balanced risk-detection profile. The dummy baseline achieved very high accuracy because most records were timely, but its balanced accuracy and ROC-AUC were 0.5000, confirming that it had no meaningful discrimination value. This result reinforces the class-imbalance treatment described above: model selection should focus on discrimination, classwise risk detection, and operational threshold trade-offs rather than overall accuracy alone.

Table 10

Selected model performance from the non-timely risk perspective.

Model	Accuracy	Balanced Accuracy	Precision Non-Timely	Recall Non-Timely	F1 Non-Timely	ROC-AUC Risk	AP Risk
Structured random forest	0.7884	0.8000	0.0518	0.8120	0.0974	0.8748	0.1181

From the non-timely perspective, the final model captured 81.20% of actual non-timely complaints at the default threshold. This is valuable for early-warning triage, but the low non-timely precision means many flagged cases will still be timely. In a complaint operation, this translates into a review queue rather than an automatic escalation decision.

Business Outcomes and Management Implications

The technical results translate into several business outcomes. The most important outcome is a practical early-warning mechanism. Instead of waiting until a complaint is close to a response deadline, the organization can score new complaints using structured metadata and route selected cases for earlier review. This can improve operational readiness, focus senior attention on potentially complex cases, and support a more proactive complaint-management culture.

A second outcome is capacity planning. Threshold analysis converts model probabilities into expected review volume. Lower thresholds catch more delayed-risk cases but create a large number of false alerts. Higher thresholds create a smaller queue with stronger alert quality but miss more delayed-risk cases. This means threshold selection is not only a data science decision; it is a management decision that should reflect team capacity, service commitments, regulatory tolerance, and risk appetite.

A third outcome is process improvement. Product and issue groups with high error rates or high non-timely risk can be reviewed by operations and product teams. The purpose is not to blame a product or company, but to identify where routing rules, investigation steps, customer communication, documentation quality, or escalation procedures may need improvement.

A fourth outcome is responsible governance. The subgroup analysis shows that model performance varies by product, company group, state, channel, year, and tags. This creates a need for model monitoring before any operational deployment. The model can support decision-making only if false alerts, missed-risk cases, subgroup stability, privacy, and drift are actively managed.

Table 11

Business outcomes from the analytical results.

Business Outcome	Evidence	Operational Use	Risk Control
Early-warning triage	Random forest ROC-AUC 0.8748; non-timely recall 0.8120	Create a risk-review queue for complaints more likely to be delayed.	Human review before escalation.
Capacity planning	Threshold table shows flagged volume at each threshold	Set alert volume according to analyst capacity and risk tolerance.	Document threshold rationale and review workload.
Product workflow review	Product error concentration and RQ1 association	Review high-error products such as payday/personal loans, bank account/service, vehicle loans, prepaid cards, and mortgage.	Do not treat product as causal proof.
Issue routing improvement	RQ2 association and issue-level error concentration	Improve issue-level routing, specialist handling, and response playbooks.	Pair model output with SME review.
Company/state monitoring	RQ4 ROC-AUC improvement of 0.1479	Use company and geography signals for internal monitoring and governance review.	Avoid direct public rankings.
Narrative-supported triage	RQ3 mixed but useful text results	Use narrative terms and length as complexity signals.	Apply privacy and fairness controls.
Relief-case prioritization	Relief-provided recall 0.7606 in response model	Flag complaints more likely to involve remediation or relief for analyst review.	Simplify target before operational use.

Threshold Analysis

Threshold analysis shows that the model can be configured for different management objectives. A broad risk-capture strategy uses a lower non-timely risk threshold and accepts more false alerts. A high-confidence queue uses a higher threshold and accepts more missed non-timely cases. Detailed threshold tables and the threshold trade-off figure are moved to Appendix B and Appendix C.

Feature Interpretation

Feature-family interpretation showed that company group accounted for 53.94% of total random forest feature importance, followed by product/sub-product at 19.70%, issue/sub-issue at 12.89%, time period at 6.49%, geography/state at 3.34%, narrative availability/length at 2.05%, submission channel at 0.80%, and customer tags at 0.79%. This reinforces the need for a multi-signal triage framework.

Company-level importance should be interpreted carefully because company variables are high-cardinality and may capture complaint volume, product mix, market presence, servicing model, or complexity. The broad Other company group had both high importance and high error concentration, making it a priority for refinement in future modeling. Detailed feature-family and coefficient tables are provided in Appendix B.

Company Response Outcome Modeling

The secondary company-response model examined whether complaint features could predict outcomes such as closed with explanation, relief provided, other response, and in progress. The target was highly imbalanced: most records were closed with explanation, while relief provided was smaller but operationally important and the other categories were very rare. The model showed useful recall for relief-provided cases, but rare categories were not stable enough for reliable standalone prediction. A simpler relief-provided versus no-relief target is recommended for future work.

Interim Limitations and Risks

The current analysis has important limitations. The CFPB dataset does not represent the full customer population. The analysis is observational and does not prove causation. Class imbalance makes accuracy misleading. Narrative availability is incomplete, and text may contain sensitive or proxy information. Company and state variables need careful interpretation because market share, complaint volume, product mix, customer base, and case complexity are not fully controlled.

Table 12

Summary limitations and mitigation actions.

Limitation Area	Meaning	Mitigation
Public complaint data	Not the full customer population or all internal complaints.	Use as analytical insight, not market-share or quality ranking.
Observational design	Associations do not prove causation.	Use cautious language and SME validation.
Class imbalance	Accuracy can hide poor minority-class detection.	Use balanced and classwise metrics.
Low non-timely precision	Many flagged complaints may still be timely.	Use human review and threshold governance.
Narrative sensitivity	Text may include sensitive or proxy information.	Use privacy, fairness, and data-minimization controls.
Company/state interpretation	Signals may reflect volume and mix, not direct performance.	Avoid direct rankings without additional controls.
Subgroup variation	Model stability differs by segment.	Monitor subgroup metrics before deployment.

The full limitations table is provided in Appendix B. The main implication is that the model should be used as decision support with human oversight, not as an automated complaint decision system.

Next Steps (for Final Report)

- Refine the broad Other company group into more meaningful categories where sample size permits.
- Select and document a final operating threshold based on analyst capacity, false-alert tolerance, and missed-risk tolerance.
- Review high-error product and issue groups with complaint operations subject-matter experts.
- Evaluate a simplified relief-provided versus no-relief outcome model for company response prediction.
- Add repository screenshots, output evidence, and final report artifacts to GitHub.
- Prepare final governance documentation covering feature exclusions, privacy, subgroup monitoring, drift review, and human-in-the-loop controls.

Business Value and Governance

The business value of this capstone lies in decision support. The model can help prioritize complaint-review queues, support early-warning dashboards, guide product-level and issue-level operational reviews, and inform governance reporting. The strongest business value comes from combining multiple signals rather than relying on a single field.

Responsible deployment would require a controlled pilot. The pilot should include a dashboard that displays risk scores, thresholds, predicted risk categories, product and issue groups, company group, state, channel, year, and narrative indicators. Users should see model scores as support indicators, not as final decisions. All escalations should remain under human ownership.

Governance should include threshold approval, drift monitoring, subgroup performance review, privacy controls for narrative text, documentation of excluded leakage fields, audit evidence, and periodic model retraining. The implementation roadmap has been moved to Appendix B to keep the main report focused on the evidence and business outcomes.

The expected business outcome is a more proactive complaint-management process. Instead of waiting for complaints to age into delayed-response cases, complaint leaders can use predicted risk scores to identify cases that may need earlier review. This can improve workload planning, reduce last-minute escalations, and support more consistent follow-up for complex product or issue categories.

The analysis also supports a practical management dashboard. The dashboard should not only show model scores; it should show why a segment is being monitored, including product, issue, company group, geography, year, narrative availability, and subgroup performance. This makes the model useful for governance discussion, not just technical prediction.

Conclusion

This interim report demonstrates that machine learning can convert large-scale public complaint data into useful decision-support insight for retail banking complaint management. The project moved beyond descriptive reporting by building a reproducible analytical pipeline, testing four research questions, comparing multiple models, and interpreting the results through both technical and business lenses.

The first major conclusion is that complaint response risk is not evenly distributed across the complaint population. Product type, issue category, company group, geography, narrative characteristics, submission channel, customer tags, and time period all provide some level of signal. However, no single variable explains response timeliness or response outcome by itself. The strongest practical value comes from combining these signals into an integrated risk-based triage framework.

The second conclusion is that model evaluation must account for class imbalance. Since timely responses dominate the dataset, simple accuracy can create a false sense of model quality. The dummy baseline confirmed this concern because it achieved high accuracy while offering no real discrimination between timely and non-timely complaints. The selected structured random forest provided stronger ROC-AUC and balanced accuracy, making it more appropriate for risk ranking and early-warning use.

The third conclusion is that customer narratives are valuable, but they require careful interpretation. Narrative text improved several overall metrics when TF-IDF features were added, showing that customer-written descriptions contain operational context not fully captured in structured fields. At the same time, narrative features did not improve every minority-class metric. Therefore, narrative analytics should support complaint triage and analyst review rather than replace structured controls or human judgment.

The fourth conclusion is that company and state variables add meaningful predictive value, especially through the improvement in balanced accuracy and ROC-AUC after those variables were added to the model. Company-level signals were particularly strong, but they must be interpreted cautiously. The CFPB data does not fully control for complaint volume, market share, customer base, servicing model, product mix, or case complexity. As a result, company-level results should be used for internal monitoring and governance review, not for public ranking or causal claims.

The business outcome of the project is a practical early-warning concept for complaint operations. A controlled pilot could use the model to flag complaints for earlier review, support product and issue monitoring, guide workload planning, and strengthen governance reporting. The model is especially useful as a triage aid when paired with thresholds, subgroup monitoring, and clear review ownership. It should not be used to automatically determine complaint outcomes or customer treatment.

The final conclusion is that responsible implementation matters as much as model performance. Before operational deployment, the model would require threshold approval, false-alert monitoring, missed-risk review, drift testing, privacy controls for narrative text, and periodic retraining. Human-in-the-loop review is necessary because the model identifies predictive associations, not definitive causes. With these controls, the project provides a defensible foundation for using analytics to improve complaint-resolution oversight in retail banking.

References

Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and machine learning: Limitations and opportunities. MIT Press.

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. Journal of Machine Learning Research, 3, 993-1022.

Breiman, L. (2001). Random forests. Machine Learning, 45, 5-32. https://doi.org/10.1023/A:1010933404324

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research, 16, 321-357. https://doi.org/10.1613/jair.953

Consumer Financial Protection Bureau. (n.d.-a). Consumer Complaint Database. https://www.consumerfinance.gov/data-research/consumer-complaints/

Consumer Financial Protection Bureau. (n.d.-b). How we share complaint data. https://www.consumerfinance.gov/complaint/data-use/

CFPB Open Tech. (n.d.). Consumer Complaint Database API field reference. https://cfpb.github.io/api/ccdb/fields.html

Data.gov. (n.d.). Consumer Complaint Database. https://catalog.data.gov/dataset/consumer-complaint-database

Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In Proceedings of the 23rd International Conference on Machine Learning (pp. 233-240). https://doi.org/10.1145/1143844.1143874

Homburg, C., & Fürst, A. (2005). How organizational complaint handling drives customer loyalty: An analysis of the mechanistic and the organic approach. Journal of Marketing, 69(3), 95-114. https://doi.org/10.1509/jmkg.69.3.95.66367

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). Applied logistic regression (3rd ed.). Wiley.

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data, 6(4), 1-21. https://doi.org/10.1145/2382577.2382579

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. The Journal of Finance, 66(1), 35-65. https://doi.org/10.1111/j.1540-6261.2010.01625.x

Maxham, J. G., III, & Netemeyer, R. G. (2002). Modeling customer perceptions of complaint handling over time: The effects of perceived justice on satisfaction and intent. Journal of Retailing, 78(4), 239-252. https://doi.org/10.1016/S0022-4359(02)00114-1

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. ACM Computing Surveys, 54(6), 1-35. https://doi.org/10.1145/3457607

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD International

Conference on Knowledge Discovery and Data Mining (pp. 1135-1144). https://doi.org/10.1145/2939672.2939778

Tax, S. S., Brown, S. W., & Chandrashekaran, M. (1998). Customer evaluations of service complaint experiences: Implications for relationship marketing. Journal of Marketing, 62(2), 60-76. https://doi.org/10.1177/002224299806200205

Vaishnav, D., Neethinayagam, M., Khaire, A., & Woo, J. (2024). Predictive analysis of CFPB consumer complaints using machine learning. arXiv. https://arxiv.org/abs/2407.06399

Appendix A: Data Dictionary

Appendix A provides the detailed data dictionary generated from the extraction notebook. The main report summarizes the dataset to preserve readability, while the detailed variable list is retained here for review.

Table A1

Data dictionary for processed complaint dataset.

Variable Name	Definition	Datatype	Handling Notes
date_received	Parsed date when CFPB received complaint	datetime	Used for year/month/quarter features.
date_received_raw	Original raw date received value	string	Preserved for auditability.
product	Financial product category	categorical	Primary RQ1 product signal.
sub_product	Detailed product subtype	categorical	Supports product-level modeling.
issue	Complaint issue category	categorical	Primary RQ2 issue signal.
sub_issue	Detailed issue category	categorical	Supports routing and complexity analysis.
consumer_complaint_narrative	Customer narrative text where available	text	Used for narrative coverage and TF-IDF model.
company_public_response	Company public response	categorical/text	Excluded from timely-response prediction to avoid leakage.
company	Company named in complaint	categorical	Grouped for model features.
state	Consumer state	categorical	Grouped geography signal.
zip_code	ZIP code	string	Not used in final model due privacy/granularity concerns.
tags	Customer tags	categorical	Older American / Servicemember / Unknown.
submitted_via	Submission channel	categorical	Web, phone, referral, postal mail, fax.
date_sent_to_company	Date complaint sent to company	datetime	Used for lag diagnostics only.
date_sent_to_company_raw	Original raw date sent value	string	Preserved for auditability.
company_response_to_consumer	Company response category	categorical	Used for response outcome analysis; excluded from timely prediction.
timely_response	Original timely response value	categorical	Source for binary target.
complaint_id	CFPB complaint identifier	string/integer	Unique complaint identifier.
year_received	Year of complaint receipt	integer/categorical	Time feature for model.
month_received	Month of complaint receipt	period/string	Used in EDA trends.
quarter_received	Quarter of complaint receipt	period/string	Used in EDA trends.
timely_response_binary	Binary target	0/1	1 = timely, 0 = non-timely.

Variable Name	Definition	Datatype	Handling Notes
has_narrative	Narrative availability flag	0/1	1 if narrative present.
narrative_length	Number of characters in narrative	numeric	Proxy for detail/complexity.
received_to_sent_lag_days	Days from received to sent to company	numeric	Process diagnostic; not core intake prediction.

Appendix B: Secondary Tables

Appendix B contains secondary tables that support the main findings. These tables were moved from the main body to keep the report more executive-focused while preserving analytical evidence.

Table B1

RQ3 structured versus structured-plus-TF-IDF model comparison.

Metric	Structured Logistic	Structured + TF-IDF	Difference
accuracy	0.7333	0.8112	0.0779
balanced_accuracy	0.7908	0.7384	-0.0524
f1_timely	0.8440	0.8947	0.0507
recall_timely	0.7316	0.8133	0.0817
precision_non_timely	0.0432	0.0483	0.0051
recall_non_timely	0.8499	0.6635	-0.1864
f1_non_timely	0.0823	0.0900	0.0077
roc_auc_non_timely_risk	0.8506	0.8357	-0.0149
average_precision_non_timely_risk	0.0666	0.0627	-0.0039

Note. Narrative features improved some overall metrics but reduced non-timely recall and some risk-sensitive metrics.

Table B2

RQ4 incremental value of company and state features.

Model	Accuracy	Balanced Accuracy	F1	ROC-AUC	Interpretation
Base: complaint characteristics	0.7162	0.6490	0.8331	0.7027	
Extended: plus company and state	0.7333	0.7908	0.8440	0.8506	
Difference: extended minus base	0.0170	0.1418	0.0109	0.1479	Company/state features add predictive value.

Table B3

Non-timely risk threshold trade-off table.

Risk Threshold	Flagged Records	Flagged %	Precision	Recall	F1	Balanced Accuracy

Risk Threshold	Flagged Records	Flagged %	Precision	Recall	F1	Balanced Accuracy
0.10	37,843	84.10%	0.0167	0.9968	0.0328	0.5791
0.15	31,304	69.56%	0.0201	0.9937	0.0394	0.6511
0.20	25,944	57.65%	0.0241	0.9889	0.0471	0.7091
0.25	20,967	46.59%	0.0296	0.9795	0.0574	0.7604
0.30	17,115	38.03%	0.0359	0.9700	0.0692	0.7990
0.35	14,530	32.29%	0.0413	0.9479	0.0791	0.8169
0.40	12,652	28.12%	0.0463	0.9258	0.0882	0.8269
0.45	11,250	25.00%	0.0499	0.8863	0.0944	0.8227
0.50	9,917	22.04%	0.0518	0.8120	0.0974	0.8000
0.55	8,449	18.78%	0.0541	0.7220	0.1006	0.7709
0.60	6,942	15.43%	0.0568	0.6224	0.1040	0.7374
0.65	4,267	9.48%	0.0701	0.4724	0.1220	0.6915
0.70	2,660	5.91%	0.0808	0.3397	0.1306	0.6423
0.75	1,478	3.28%	0.1035	0.2417	0.1450	0.6059
0.80	341	0.76%	0.2111	0.1137	0.1478	0.5538
0.85	81	0.18%	0.4198	0.0537	0.0952	0.5263
0.90	11	0.02%	0.8182	0.0142	0.0280	0.5071

Table B4

Random forest feature-family importance.

Feature Family	Total Importance	Feature Count	Top Feature	Importance %
Company group	0.5394	51	company grouped Other	53.94%
Product / sub-product	0.1970	56	sub product General-purpose credit card or cha...	19.70%
Issue / sub-issue	0.1289	183	sub issue Unknown	12.89%
Time period	0.0649	9	year received 2017	6.49%
Geography / state	0.0334	51	state grouped CA	3.34%
Narrative availability / length	0.0205	2	narrative length	2.05%
Submission channel	0.0080	6	submitted via Web	0.80%
Customer tags	0.0079	4	tags Unknown	0.79%

Table B5

Company response group distribution.

Company Response Group	Count	Percent
Closed with explanation	1,038,396	80.56%
Relief provided	247,704	19.22%
Other response	2,844	0.22%
In progress	22	0.00%

Table B6

Company response outcome classification report.

Class	Precision	Recall	F1-score	Support
Closed with explanation	0.9194	0.5515	0.6895	36252.0
In progress	0.0000	0.0000	0.0000	1.0
Other response	0.0151	0.7778	0.0296	99.0
Relief provided	0.3628	0.7606	0.4913	8648.0
accuracy	0.5922	0.5922	0.5922	0.5922
macro avg	0.3243	0.5225	0.3026	45000.0
weighted avg	0.8104	0.5922	0.6499	45000.0

Table B7

Detailed limitations and mitigation actions.

Limitation	Meaning	Business Impact	Mitigation
Public complaint data	CFPB complaints are not the full customer population or all internal complaints.	Results should not be interpreted as complete customer experience or company quality.	Use findings for insight and triage design, not market-share or quality rankings.
Observational design	The models identify associations and predictive patterns, not causal effects.	Users may overinterpret model outputs as reasons for delay.	Use cautious language and validate findings with subject-matter experts.
Class imbalance	Timely responses dominate while non-timely complaints are rare.	Accuracy can look strong even when risk detection is weak.	Use balanced accuracy, ROC-AUC, PR-AUC, classwise metrics, and thresholds.
Low non-timely precision	The model captures many delayed-risk cases but flags many timely cases.	Complaint teams may face a false-alert workload.	Use human review and set thresholds based on capacity and risk tolerance.
Threshold choice	Different thresholds produce different recall and precision trade-offs.	Low thresholds create workload; high thresholds miss risk cases.	Document flagged volume, precision, recall, F1, and review-capacity assumptions.
Incomplete narratives	Not every complaint includes consumer narrative text.	Text findings may not represent all complaints equally.	Report coverage and use narratives as supporting signals only.
Sensitive text signals	Narratives may include hardship, identity, location, fraud, or other proxy information.	Text analytics may create privacy or fairness concerns.	Apply data minimization, privacy review, and responsible AI controls.
Company/state interpretation	Company and state patterns may reflect volume, mix, servicing model, or case complexity.	Signals could be misread as direct rankings.	Use internally for monitoring and avoid causal or public ranking claims.
Broad Other company group	The Other group combines many smaller or less frequent companies.	The model may over-flag a mixed group with unstable patterns.	Refine grouping into banks, fintechs, servicers, credit unions, and low-volume firms.
High-cardinality fields	Company, issue, sub-issue, and state have many categories.	Random forest importance may overstate some feature families.	Use feature-family interpretation, coefficients, subgroup review, and SME validation.

Limitation	Meaning	Business Impact	Mitigation
Subgroup variation	Performance differs by product, company, state, channel, year, and tags.	Some groups may receive more false alerts or weaker risk detection.	Monitor subgroup precision, recall, false positives, and false negatives.
Response outcome imbalance	Most responses close with explanation; rare categories are difficult to model.	Other response and In progress are not stable standalone targets.	Consider relief-provided versus no-relief or merge rare categories.
Leakage risk	Post-response fields may not be available at intake time.	Including them could inflate operational performance.	Exclude company_response_to_consumer and company_public_response for timely prediction.
Drift over time	Complaint patterns and operations can change across years.	Historical performance may not hold in future periods.	Track drift, validate on recent data, and retrain periodically.
Limited transferability	The model uses U.S. CFPB public complaint data.	It may not transfer directly to another bank or jurisdiction.	Validate locally using internal data, local products, and local regulations.
No automated decisions	The model predicts risk patterns, not final complaint outcomes.	Automated action could create fairness or compliance risk.	Use decision support with human-in-the-loop review and audit evidence.

Table B8

Detailed implementation roadmap.

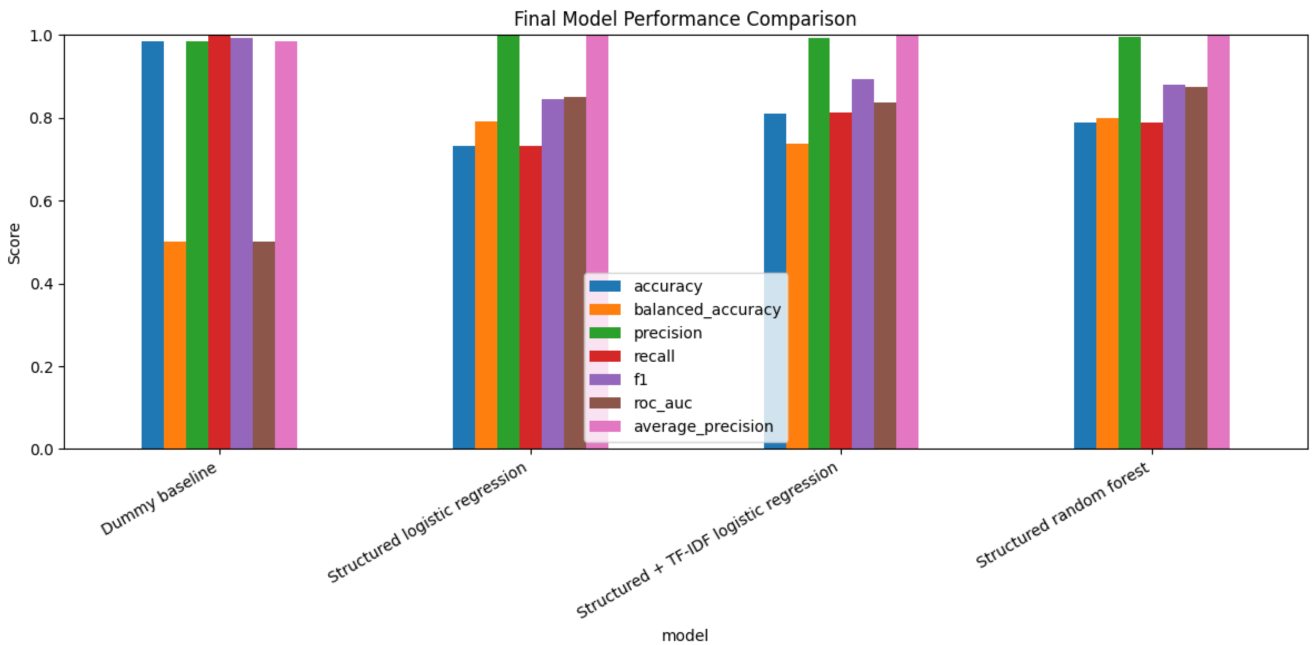
Phase	Focus	Activities	Owner	Risk Control
Phase 1	Analytical validation and model readiness	Validate final model results, ROC-AUC, balanced accuracy, classwise metrics, feature importance, subgroup performance, and error concentration.	Analytics with complaint operations	Document imbalance, leakage controls, assumptions, and why accuracy alone was not used.
Phase 2	Business validation and operational review	Review high-risk product groups, issue groups, company groups, state patterns, and year-level trends with SMEs.	Complaint operations, analytics, risk, product owners	Avoid causal interpretation or direct company rankings.
Phase 3	Threshold calibration	Select non-timely risk threshold based on capacity, false-alert tolerance, missed-risk tolerance, and escalation policy.	Complaint operations, analytics, risk, compliance	Document threshold rationale and workload impact.
Phase 4	Pilot dashboard and monitoring view	Build dashboard for risk scores, thresholds, product, issue, company group, state, channel, narrative indicators, and top segments.	Operations analytics, complaint management, technology	Label predictions as decision-support only.
Phase 5	Workflow integration and human review	Route selected high-risk complaints for earlier review, supervisor monitoring, specialist handling, or escalation.	Complaint operations, technology, risk, compliance	Keep human-in-the-loop controls; no automated adverse treatment.
Phase 6	Model governance and responsible AI controls	Monitor drift, subgroup performance, threshold performance, product errors, issue errors, company group stability, FP/FN rates.	Model governance, risk, compliance, analytics, business owners	Maintain audit trail, privacy controls, approvals, and retraining schedule.
Phase 7	Future enhancement	Refine company grouping, test thresholds, improve rare-class handling, evaluate relief-vs-no-relief model, and explore improved text features.	Analytics with operations and governance	Validate enhancements before adoption using risk-sensitive metrics and workload analysis.

Appendix C: Supporting Figures

Appendix C contains the detailed figures and charts moved out of the main report. The main text references these figures where they support the conclusions.

Figure C1

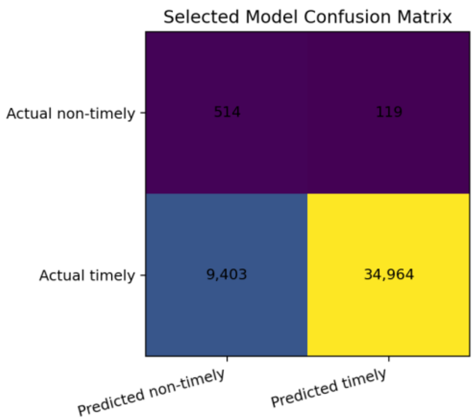
Final model performance comparison.



Note. The random forest offers the strongest ROC-AUC and balanced risk-detection profile; the dummy baseline demonstrates why accuracy is misleading.

Figure C2

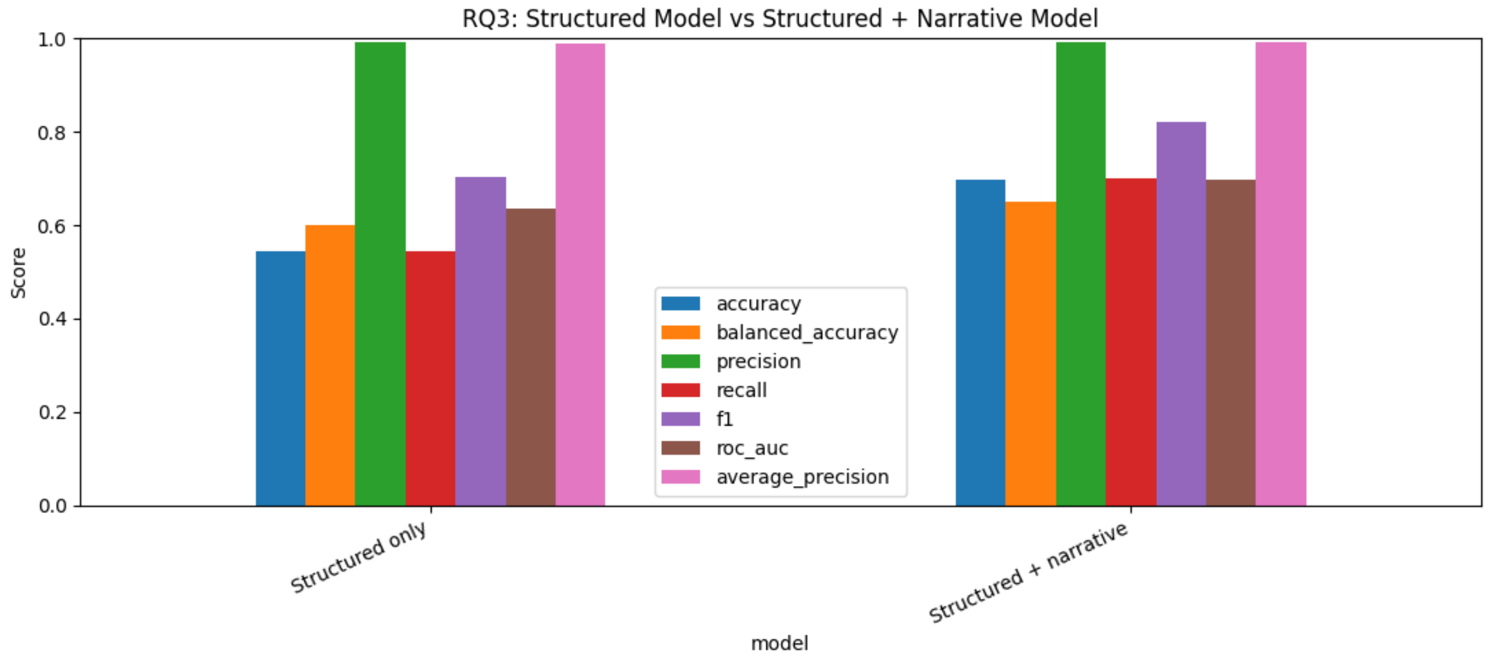
Confusion matrix for selected structured random forest.



Note. The selected model catches many non-timely complaints but creates a false-alert workload.

Figure C3

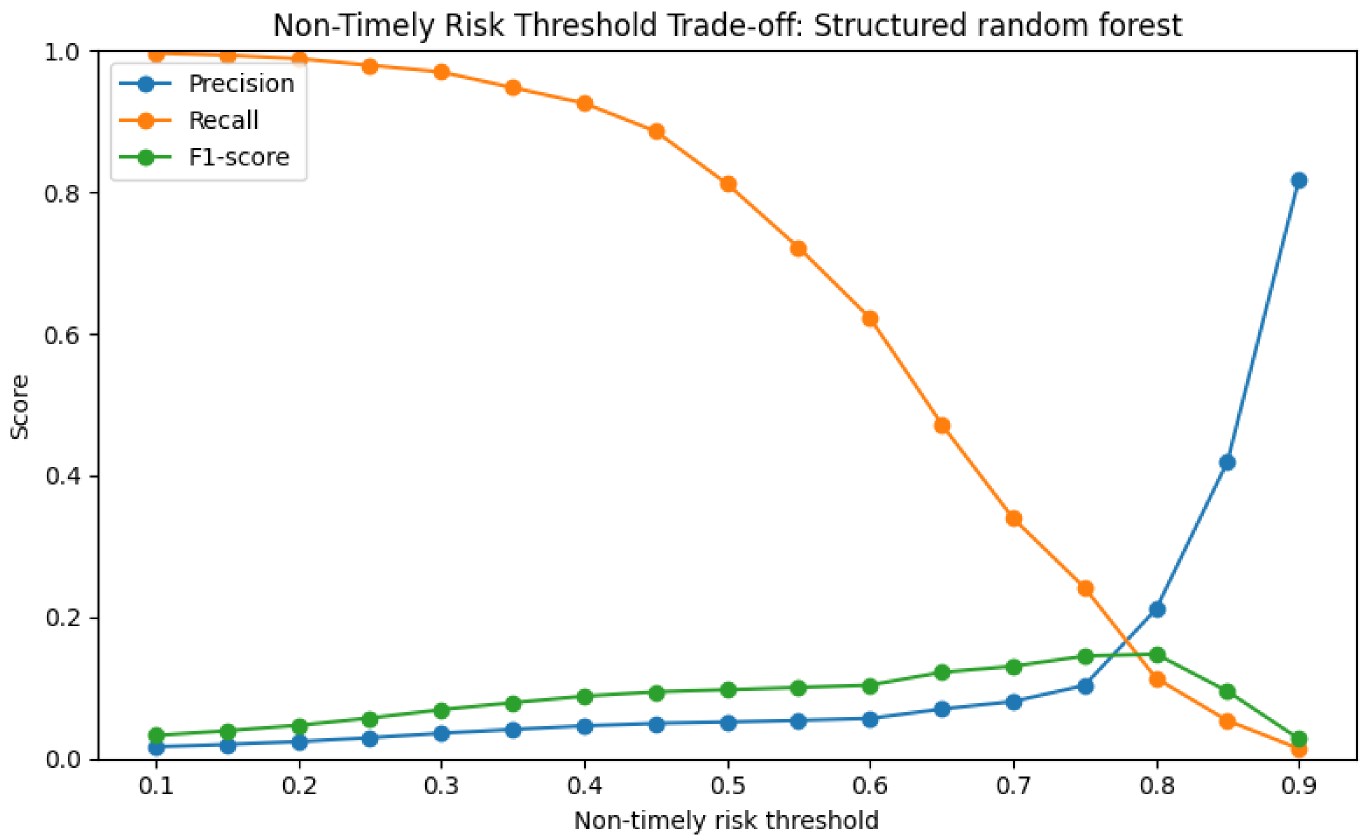
RQ3 structured model versus structured plus narrative model.



Note. Text features improve some metrics but do not uniformly improve minority-class risk detection.

Figure C4

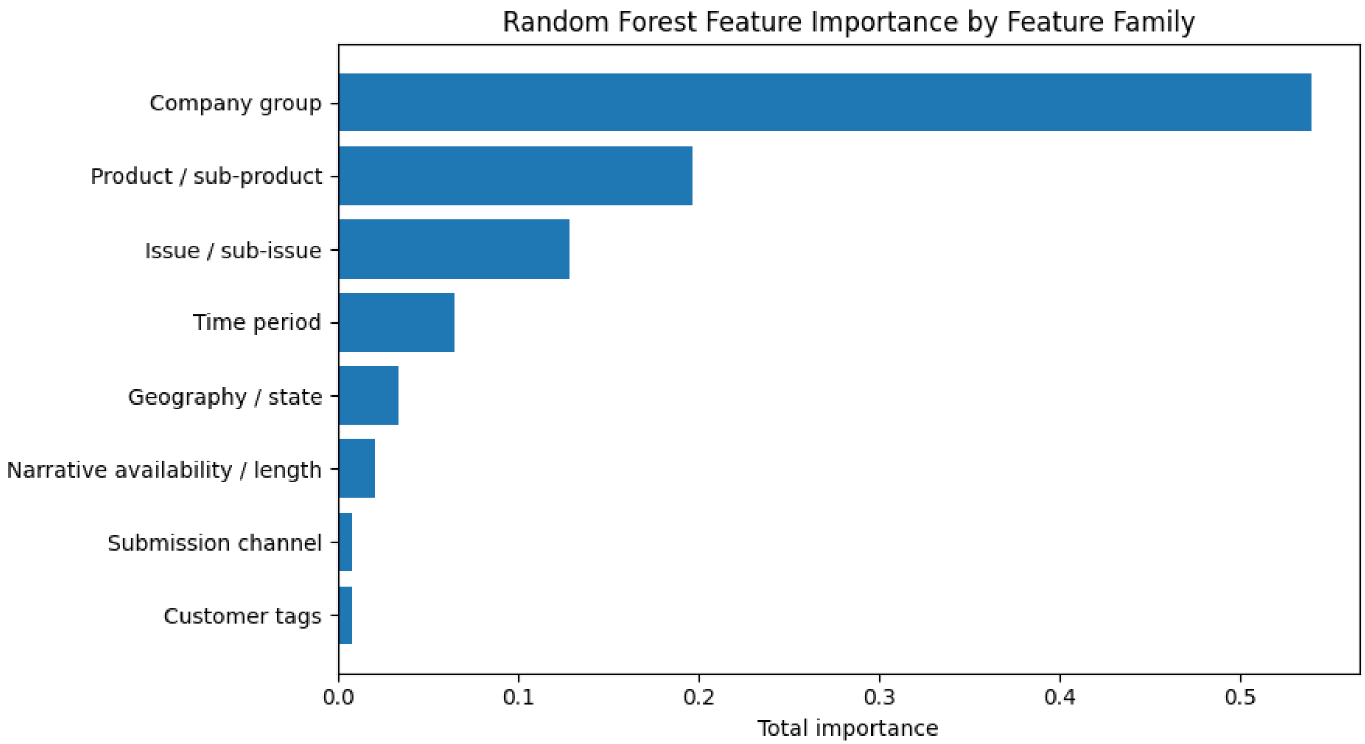
Non-timely risk threshold trade-off.



Note. Threshold selection controls the trade-off between risk coverage and alert quality.

Figure C5

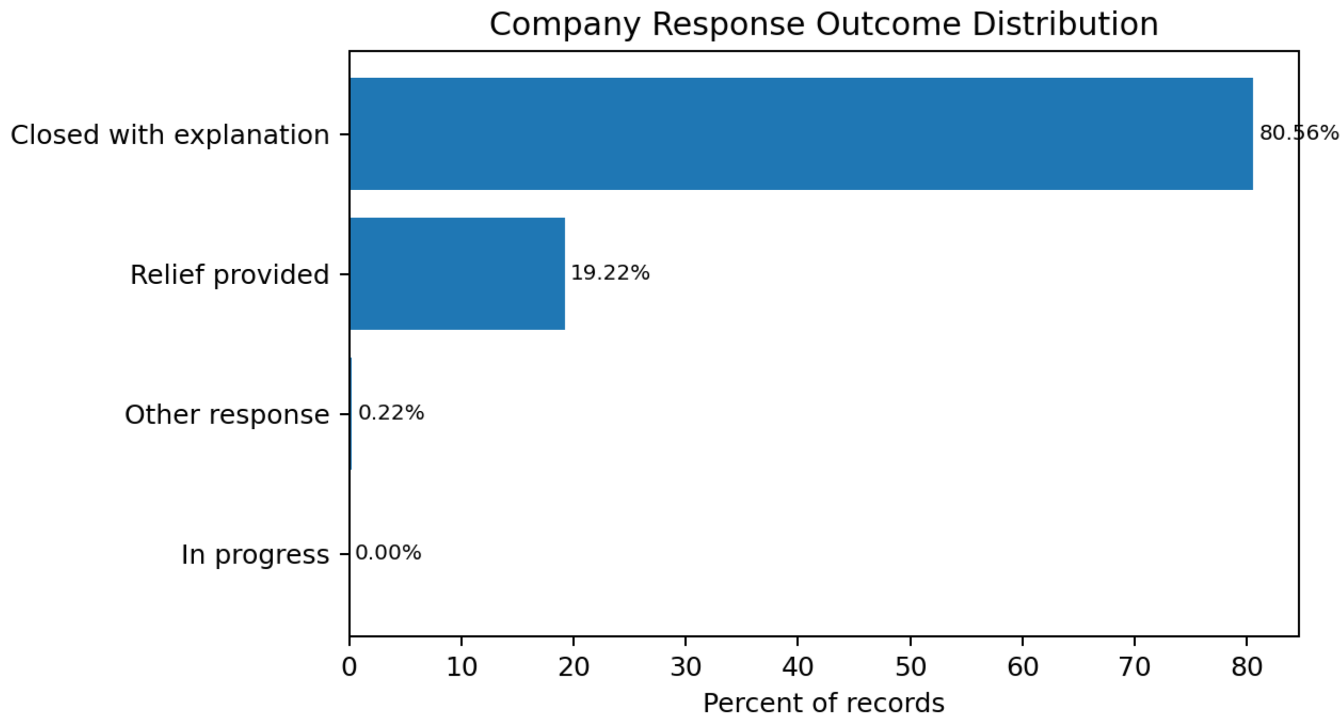
Random forest feature importance by feature family.



Note. Company group, product/sub-product, and issue/sub-issue dominate the final model.

Figure C6

Company response outcome distribution.



Note. Most complaints close with explanation, while relief provided is the main actionable secondary outcome.

Appendix D: Notebook and Output Evidence

The report is based on five notebooks uploaded for the capstone workflow. Notebook 01 created the processed dataset and data dictionary. Notebook 02 generated EDA figures and tables. Notebook 03 conducted hypothesis testing. Notebook 04 trained and compared models. Notebook 05 interpreted the final models and generated business findings, limitations, and governance outputs.

Table D1

Notebook workflow evidence.

Notebook	Role in Report
01_data_extraction_colab_csv_v2.ipynb	Raw CSV validation, chunk extraction, product/date filtering, feature creation, data dictionary, processed output.
02_exploratory_analysis_v2.ipynb	EDA tables and visual interpretation for product, issue, company, state, channel, timely response, company response, narratives, and trends.
03_hypothesis_tests_v2.ipynb	Hypothesis testing for RQ1-RQ4, including chi-square tests, effect sizes, and final interpretation.
04_model_training_v2.ipynb	Model training, baseline comparison, logistic regression, TF-IDF model, random forest, classwise metrics, and model selection.
05_model_interpretation_v2.ipynb	Threshold analysis, error analysis, subgroup performance, feature-family importance, company response outcome modeling, limitations, roadmap, and final summary.